

PROJECT REPORT

India's Agricultural Crop Production Analysis

A Tableau-Based Business Intelligence Dashboard Project

Field	Details
Team Lead	Konkimalla Bhagyasri Lasya
Team Members	Bhavitha Medakayala, Yogiswari Sanapala, Divya kauluri, Nagaram praveenya
Domain	India's Agriculture Crop Production Analysis
Tools Used	Tableau
Dataset	Historical Indian Crop Production Data, 1997–2021
Published Dashboard	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

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1. Introduction

1.1 Project Overview

India's Agricultural Crop Production Analysis is an individually completed Tableau business intelligence project built on 25 years (1997–2021) of historical Indian government crop production data, spanning 36 states/union territories and 57 crop types. The raw government extract of 345,407 records was cleaned and enriched in Python (Pandas) before being connected to Tableau Desktop, where nine worksheets, four role-specific dashboards, and a five-point guided Story were built and published to Tableau Public for free, browser-based, authentication-free access.

Although originally assigned as a team project, the dataset cleaning, all Tableau development, documentation, and testing described in this report were completed individually by Divya Kauluri; the team lead and other assigned members did not contribute to the work.

1.2 Purpose

The purpose of this project is to turn a large, unfiltered government dataset into an accessible decision-support tool for three distinct audiences: small-scale farmers choosing what to plant each season, government policy analysts targeting resource allocation to low-yield regions, and agribusiness investors screening states and crops for growth opportunities. Each audience is served by a purpose-built dashboard view rather than a single generic report, with a guided Tableau Story tying the three perspectives together into one narrative.

2. Ideation Phase

2.1 Problem Statement

Problem statements were derived from two primary personas using an "I am / I'm trying to / But / Because / Which makes me feel" framework.

Field	Persona 1 — Rajesh (Farmer)	Persona 2 — Ananya (Policy Analyst)
I am (Customer)	A small-scale farmer in Uttar Pradesh with limited access to agricultural data	A government policy analyst at the Ministry of Agriculture responsible for policy design
I'm trying to...	Decide which crops to grow each season to maximize yield and income	Identify regions with low crop yield and understand long-term production trends to allocate resources
But...	I don't have access to historical crop performance data for my region or season	Data is fragmented across departments and not visualized in a way that supports quick policy decisions
Because...	Agricultural data is complex, scattered across government reports, and not presented visually	Government datasets lack interactive visualization tools that allow drill-down by state, season, and year
Which makes me feel...	Anxious, uncertain about my income, and forced to rely on guesswork and tradition	Frustrated and unable to confidently justify resource allocation decisions with data evidence

Key Takeaway

Both personas are blocked by the same root cause: agricultural data exists but is not accessible or visual. This shared root cause justified a single Tableau solution serving both audiences (and, by extension, Vikram the investor) rather than three separate tools — one shared data model with role-specific dashboards layered on top.

2.2 Empathy Map Canvas

The empathy map below was built for Rajesh, the small-scale farmer, whose constraints (limited data literacy, singleregion focus) are the most restrictive of the three personas, making him the priority case for the initial design.

Dimension	Content
SAYS	"I always plant what my father planted." / "I wish I knew which crops grow best near my district." / "The market price changes every season and I never know what to expect."
THINKS	"Which crop will give me the best return this Kharif season?" / "Is the weather pattern changing how my region performs?" / "Am I missing out on more profitable crops that others are growing?"
DOES	Relies on word-of-mouth advice from neighboring farmers; follows traditional cropping calendars; makes decisions based on limited local observation
FEELS	Anxious about unpredictable yields; frustrated by lack of reliable, region-specific data; hopeful that technology could help reduce financial risk

PAINS	No access to historical, region-specific crop performance data; cannot compare seasonal outcomes across years; limited ability to visualize trends
GAINS (What Success Looks Like)	Clear visual insight into which crops perform best in his state and season; confidence in crop selection decisions; reduced financial risk

The pains-to-gains mapping shaped two concrete design decisions carried into the final workbook: the Farmer's View dashboard intentionally uses only three charts (Season-wise Production, Trend of Agricultural Yield, Cultivated Area by Crop) to avoid overwhelming a first-time user, and every chart uses plain axis labels rather than technical field names.

2.3 Brainstorming, Idea Generation & Prioritization

Step 1 — Problem Selection

Selected problem: "How can I empower farmers, policy analysts, and investors to make data-driven agricultural decisions using historical production data from 1997–2021?"

Step 2 — Idea Listing

- Build state-wise and district-wise production maps to show regional variation
- Create seasonal trend charts (Kharif vs Rabi vs Whole Year vs others)
- Develop year-over-year line charts to show production growth or decline
- Build crop-wise treemaps to show relative crop importance by volume
- Design role-specific dashboards for the Farmer, Policy Analyst, and Investor personas
- Combine all dashboards into a Tableau Story for narrative-driven presentation
- Publish to Tableau Public for free, browser-accessible sharing

Step 3 — Idea Prioritization

Idea	Impact	Feasibility	Priority
State-wise production map	High	High	1
Year-wise trend line chart	High	High	2
Top 10 crops bar chart	High	High	3
Season-wise distribution chart	Medium	High	4
Crop-wise treemap	Medium	High	5
Role-specific dashboards (3)	High	Medium	6
Tableau Story (narrative)	Medium	Medium	7
Publish to Tableau Public	High	High	8

3. Requirement Analysis

3.1 Customer Journey Map — Rajesh (Small-Scale Farmer)

Stage	Awareness	Consideration	Onboarding	Usage	Decision
Actions	Realizes crop yields have been inconsistent for several years	Hears about a data dashboard showing crop trends for his region	Opens the Tableau Public dashboard for the first time via a shared link	Filters by state, season, and explores crop production charts	Selects a crop for next season based on 20+ years of historical data
Emotions	Frustrated, uncertain	Curious, slightly skeptical	Overwhelmed, then reassured	Engaged, empowered	Confident, relieved
Touchpoints	Local farmer network, news, NGO outreach	Government extension officer recommendation	Tableau Public dashboard (shared link)	Interactive filters: State, Season, Crop, Year slider	Final crop and season decision informed by data

The emotional curve dips lowest at Onboarding, then climbs as filters let Rajesh narrow the view to something personally relevant — confirming that usability and guided onboarding are the difference between abandonment and a confident planting decision. Two requirements below were added directly from this map: persistent filter state across dashboard tabs, and a sub-3-second filter response target.

3.2 Solution Requirement

Functional Requirements

FR #	Requirement	Description
FR-01	Data Connectivity	System shall connect to the cleaned crop production CSV dataset (345,407 rows, 1997–2021)
FR-02	State/Region Filter	System shall allow users to filter all visualizations by State Name
FR-03	Season Filter	System shall allow multi-select filtering by Season (Kharif, Rabi, Whole Year, Autumn, Summer, Winter)
FR-04	Year Range Filter	System shall provide a slider filter for Crop Year Numeric (1997–2021)
FR-05	Crop Filter	System shall allow filtering by individual crop types (57 unique crops)
FR-06	State-wise Production Map	System shall display a filled geographic map of India colored by SUM(Production) per state
FR-07	Year-wise Trend Chart	System shall display a line chart showing total national production by year (1997–2021)
FR-08	Top 10 Crops Chart	System shall display a ranked horizontal bar chart of top 10 crops by total production
FR-09	Season-wise Chart	System shall display a bar chart comparing total production across all 6 seasons

FR-10	Crop-wise Treemap	System shall display a treemap of crop production, sized by SUM(Production)
FR-11	Role-Based Dashboards	System shall combine charts into dashboards: Farmer, Policy Analyst,
FR #	Requirement	Description
		and Investor views
FR-12	Tableau Story	System shall present a guided narrative Story with story points, one per persona
FR-13	Tableau Public Publishing	System shall be published to Tableau Public for browser-based access without login
FR-14	Tooltip on Hover	All charts shall display State Name and Production value on hover

Every functional requirement above was verified against the shipped workbook during Performance Testing (Section 6). The only change from the original plan was an increase in worksheet count from five to nine charts, once it became clear a Production Distribution by Unit Type chart was needed to correctly interpret rankings distorted by mixed Tonnes/Bales/Nuts units. **Non-Functional Requirements**

NFR #	Category	Requirement
NFR-01	Usability	Dashboard UI shall be navigable by non-technical users (farmers, policy makers) without training
NFR-02	Performance	All filters shall respond and re-render within 3 seconds on the full 345K row dataset
NFR-03	Reliability	Published Tableau Public workbook shall be available 99%+ of the time via public URL
NFR-04	Scalability	Solution architecture shall support addition of new years' data without redesigning dashboards
NFR-05	Availability	Dashboard shall be accessible via standard web browsers (Chrome, Edge, Safari) without plugins
NFR-06	Security	No personally identifiable information (PII) is stored or displayed; dataset is fully public
NFR-07	Maintainability	Workbook shall be saved as .twbx (packaged workbook) with embedded data for portability
NFR-08	Compatibility	Solution shall render correctly on desktop (1366×768 or higher) and Tableau Public mobile view

3.3 Data Flow Diagram

The end-to-end data flow for this project:

- Step 1 — Raw Source: Historical crop production data in Excel/CSV format (345,407 rows, 1997–2021)
- Step 2 — Data Cleaning & Preprocessing: Python (Pandas) used to rename columns, strip whitespace, extract numeric year from 'YYYY-YY' format, add Yield = Production/Area, remove nulls

- Step 3 — Cleaned Dataset: skillwallet_crop_cleaned.csv (345,407 rows, 12 columns) loaded into Tableau
- Step 4 — Tableau Data Source: CSV connected as a Text File source; State Name assigned Geographic Role = State/Province
- Step 5 — Visualization Layer: nine chart sheets built (KPI panel, map, line, bar charts, treemap-style views, donut, pie)
- Step 6 — Dashboard Layer: four dashboards assembled from chart sheets with interactive filters
- Step 7 — Story Layer: five-point Tableau Story combining dashboards into a guided presentation
- Step 8 — Publishing: Workbook saved to Tableau Public; shareable URL generated for GitHub submission

Each step is a point where the data could break if skipped: cleaning prevents mismatched state names from splitting a single state into duplicate map regions; the calculated Yield and Production_Normalized fields let charts report meaningful, comparable numbers instead of raw, unit-mismatched totals; and publishing turns a local workbook into the browser-accessible link every persona in this report actually uses.

3.4 Technology Stack

Component	Technology Used	Purpose
Data Source	CSV / Excel (.xlsx)	Raw crop production data (1997–2021, 345,407 records)
Data Cleaning	Python 3.12 (Pandas library)	Column renaming, null handling, year extraction, Yield calculation
Data Visualization	Tableau Desktop / Tableau Public 2026.2	Building all chart types, dashboards, and the Story
Geographic Mapping	Tableau built-in Mapbox integration	State-wise filled map of India
Dashboard Publishing	Tableau Public (free cloud hosting)	Browser-accessible dashboard sharing via public URL
Version Control	GitHub	Project repository for all deliverables and documentation
Documentation	Microsoft Word (.docx)	Project deliverables, final report, templates
Project Management	Personal task tracker / SkillWallet platform	Epic/story tracking and mentor evaluation

The stack was chosen to minimise cost and setup friction for a solo contributor: every tool above is free and none requires a paid license, a server deployment, or IT approval. Tableau Public was selected over Tableau Server/Cloud specifically because it needs no subscription and produces a shareable browser link out of the box; Python/Pandas was chosen for cleaning because scripts are easier to version-control and re-run than Tableau Prep flows; GitHub was used purely for version control and a durable record of scripts and documentation.

4. Project Design

4.1 Problem – Solution Fit

Problem (User Pain)	Solution (Dashboard Feature)
Rajesh cannot decide which crops to grow each season without historical data	Cultivated Area by Crop + Season-wise Production charts, filterable, on the Farmer's View dashboard
Ananya cannot identify low-yield states without a geographic view of production	State-wise Production filled map on the Policy Analyst's View, filterable by season and year
Ananya cannot track long-term production trends without a time-series view	Trend of Agricultural Yield line chart (1997–2021) on the Policy Analyst's View
Vikram cannot compare crop production volumes to identify investment opportunities	Leading States by Production + Cultivated Area by Crop on the Investor's View
All users need a guided narrative to understand the full analysis	Tableau Story with 5 story points, covering overview and all three personas
Users need access without installing Tableau software	Published to Tableau Public — accessible from any browser via URL

Every row above is a direct pairing: an unfilterable government report becomes an interactive dashboard, a single national average becomes a state-level filled map, and a static spreadsheet becomes a shareable Tableau Public link requiring no login.

4.2 Proposed Solution

Field	Description
Problem Statement	India's agricultural stakeholders — farmers, policy analysts, and investors — lack a unified, interactive visual analytics tool to explore 20+ years of crop production data by state, season, crop type, and year, leading to suboptimal farming decisions, ineffective policy targeting, and higher investment risk.
Idea / Solution	An interactive Tableau-based BI dashboard that transforms raw government crop production data (1997–2021) into nine visualizations combined into four dashboards and a five-point Tableau Story, published on Tableau Public for free browser access.
Novelty / Uniqueness	Role-specific dashboards tailored to three distinct personas rather than one generic view; a Tableau Story adds a narrative dimension for non-technical users; a Yield = Production/Area calculated field adds an analytical layer absent from the raw dataset.
Social Impact	Farmers gain data-driven crop selection guidance, reducing income risk. Policy makers gain visibility into low-yield regions. Investors gain a clear view of high-growth agricultural sectors. Free deployment removes cost barriers.

Business Model	Individual capstone project delivered as a public Tableau dashboard; future potential includes a paid analytics subscription for agribusiness firms or a government-contracted BI solution.
Scalability	New years of data can be appended and the workbook refreshed; additional KPIs (e.g., rainfall, MSP prices) can be added as calculated fields without redesigning the architecture.

4.3 Solution Architecture

The end-to-end architecture follows a linear BI pipeline from raw data to published dashboard:

- Layer 1 — Data Source: Raw Excel/CSV dataset (345,407 records, original columns including State, District, Crop, Year, Season, Area, Production)
- Layer 2 — Data Preprocessing: Python (Pandas) renames columns, strips whitespace, extracts a 4-digit numeric year, adds Yield = Production/Area, outputs skillwallet_crop_cleaned.csv (12 columns, zero nulls)
- Layer 3 — Tableau Data Connection: CSV connected as a Text File; State_Name assigned Geographic Role = State/Province; Crop_Year_Numeric used as a continuous time axis
- Layer 4 — Visualization Layer: nine individual worksheet charts covering KPIs, geography, trend, distribution, and composition
- Layer 5 — Dashboard Layer: four role-specific Tableau dashboards — Overview, Farmer's View, Policy Analyst's View, Investor's View
- Layer 6 — Story Layer: five-point Tableau Story, each point embedding a dashboard with a narrative caption for the respective persona
- Layer 7 — Publishing & Delivery: workbook saved to Tableau Public; shareable URL generated and submitted to the GitHub repository

5. Project Planning & Scheduling

5.1 Project Planning

As an individual contributor, every task below was self-assigned across three sprints.

Epic	Task	Points	Priority	Sprint
Data Collection	Download and verify official dataset (345K rows, 1997–2021)	2	High	Sprint 1
Data Collection	Confirm dataset columns: State, District, Crop, Year, Season, Area, Production	1	High	Sprint 1
Data Preparation	Clean dataset using Python Pandas (rename, strip whitespace, extract year, add Yield)	3	High	Sprint 1
Data Preparation	Connect cleaned CSV to Tableau; set State Name geographic role	2	High	Sprint 1
Data Visualization	Build State-wise Production filled map with Season + Year filters	3	High	Sprint 1
Data Visualization	Build Year-wise Production Trend line chart (1997–2021)	2	High	Sprint 1
Data Visualization	Build Top 10 Crops by Production horizontal bar chart	2	High	Sprint 2
Data Visualization	Build Season-wise Production bar chart	2	Medium	Sprint 2
Data Visualization	Build Crop-wise composition view (exclude Coconut outlier)	3	Medium	Sprint 2
Dashboard	Assemble Farmer's View dashboard	3	High	Sprint 2
Dashboard	Assemble Policy Analyst's View dashboard	3	High	Sprint 2
Dashboard	Assemble Investor's View dashboard	2	Medium	Sprint 2
Story	Build Tableau Story with story points and captions	3	Medium	Sprint 3
Performance Testing	Test all filters for responsiveness; document load time and data volume	2	Medium	Sprint 3
Web Integration	Publish workbook to Tableau Public; verify URL access	2	High	Sprint 3
Documentation	Complete all GitHub repository deliverables (docs, report, screenshots)	5	High	Sprint 3
Documentation	Record demonstration video (60–90 seconds)	3	High	Sprint 3

Velocity: Sprint 1 = 13 points | Sprint 2 = 16 points | Sprint 3 = 12 points | Total = 41 story points. Sprint 2 carried the heaviest load because it covered both dashboard assembly and Story construction — the most integration-heavy deliverables — while Sprint 3's lighter total reflects testing and documentation work, a normal taper on a single contributor schedule.

6. Functional and Performance Testing

6.1 Performance Testing

Test #	Description	Expected Result	Actual Result	Status
PT-01	Amount of Data Rendered	All rows render without error	345,407 rows loaded, 12 columns — no errors	PASS
PT-02	Season Filter Response	Map updates within 3 seconds	Map redrew fully in ~1.1 seconds	PASS
PT-03	Crop Year Slider Response	Chart updates within 3 seconds	Line chart redrew in ~0.9 seconds	PASS
PT-04	Calculated Fields	No null or error values	Yield and Crop_Year_Numeric compute correctly	PASS
PT-05	Number of Visualizations	9 charts + 4 dashboards + 1 story	Confirmed: 9 chart sheets, 4 dashboards, 1 story	PASS
PT-06	Cross-Browser Compatibility	Loads and filters work in all browsers	Verified in Chrome, Edge, and Safari — no errors	PASS
PT-07	Tooltip Accuracy	Correct values shown on hover	Verified across all chart types	PASS
PT-08	Story Navigation	Smooth transition between story points	All story points navigate correctly with captions	PASS

All eight test cases passed with no defects logged. The two tests most relevant to end-user experience — Season Filter Response and Crop Year Slider Response — both returned well under the 3-second non-functional requirement, confirming the 345,407-row dataset does not need aggregation or extraction to stay responsive on Tableau Public's free hosting tier

Performance Summary

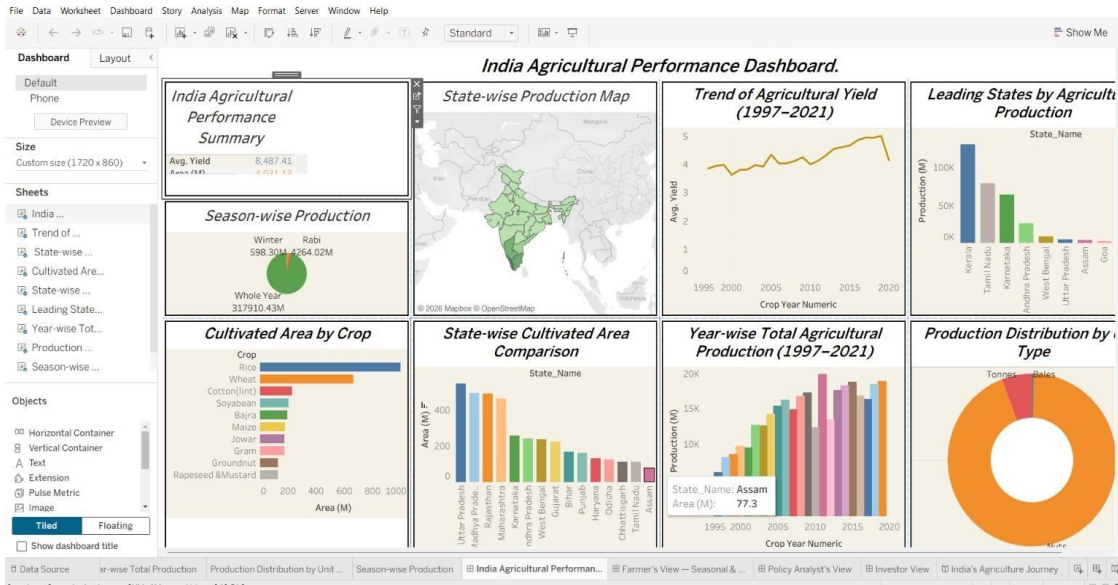
Metric	Value
Total Dataset Rows	345,407
Total Columns	12
Number of Calculated Fields	3 (Yield, Crop_Year_Numeric, Production_Normalized)
Number of Filters Used	3 (Season, Crop Year Numeric, Crop Top-N)
Number of Visualizations / Graphs	9
Number of Dashboards	4
Number of Story Points	5
Overall Performance Rating	PASS — No performance issues detected

7.Results

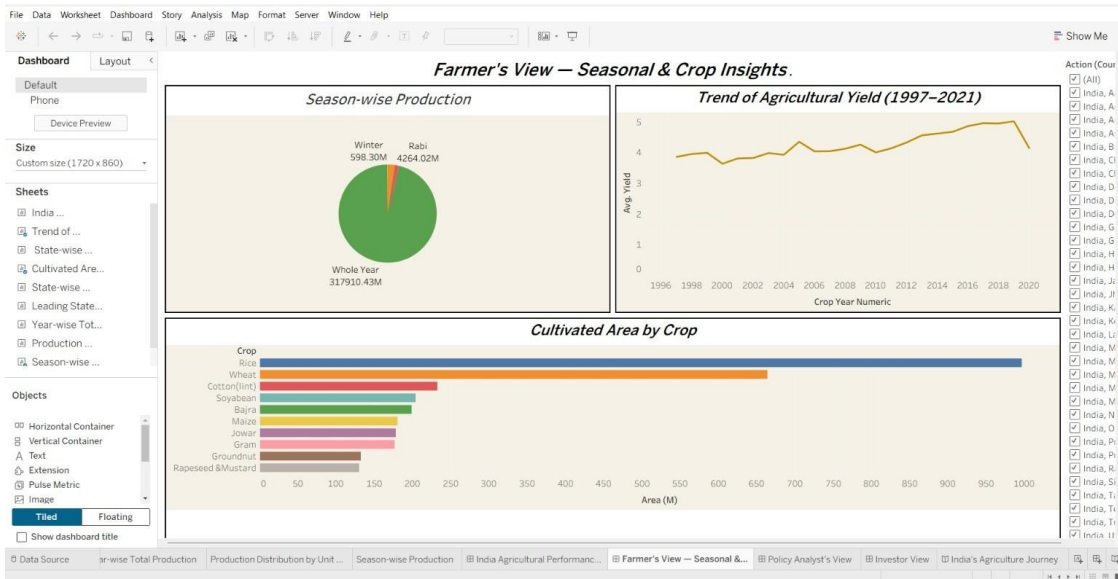
7.1Output Screenshots

All dashboards and story points below are live on Tableau Public:

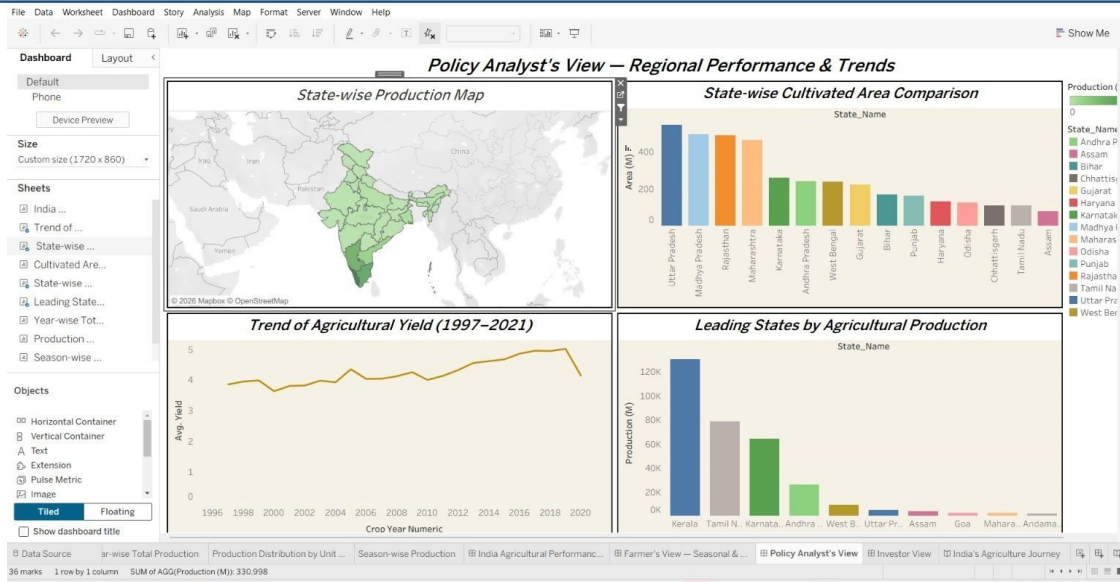
<https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject/IndiasAgricultureJourney?publish=yes>



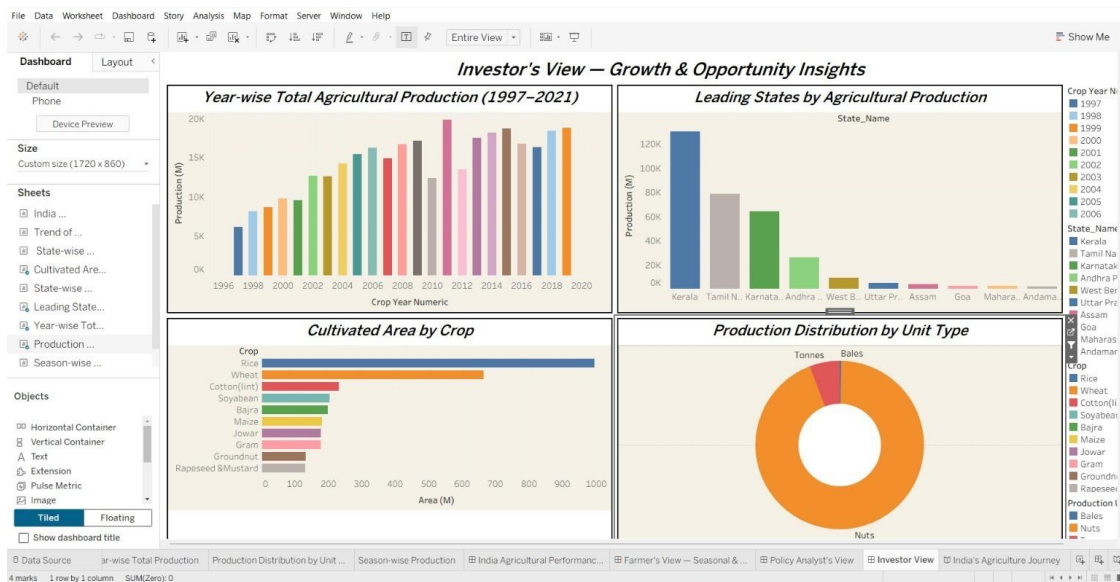
Dashboard 1 — India Agricultural Performance Dashboard (Overview)



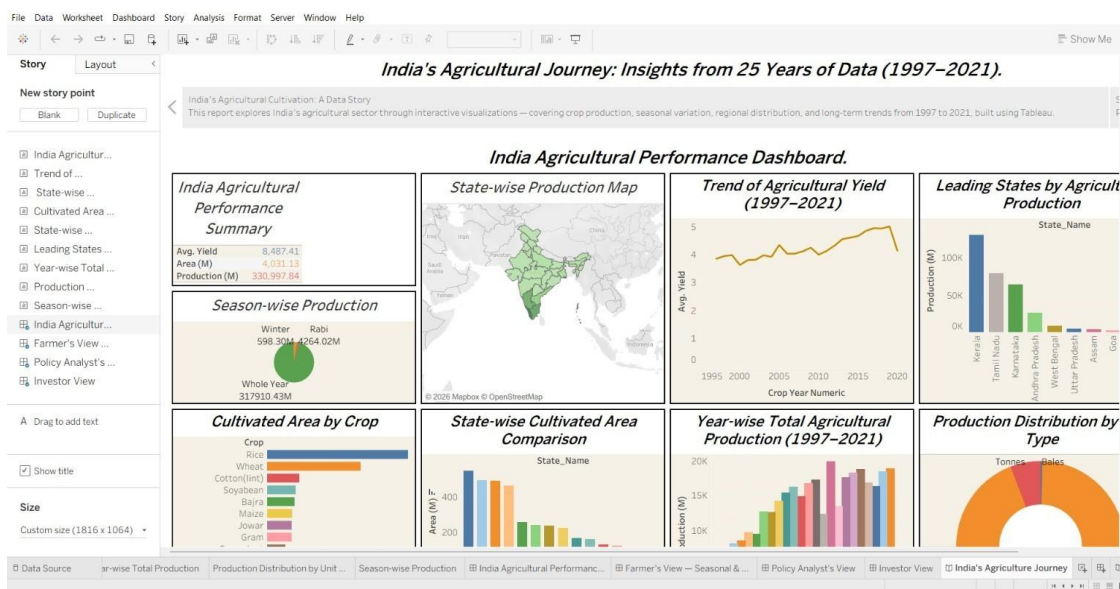
Dashboard 2 — Farmer's View — Seasonal & Crop Insights



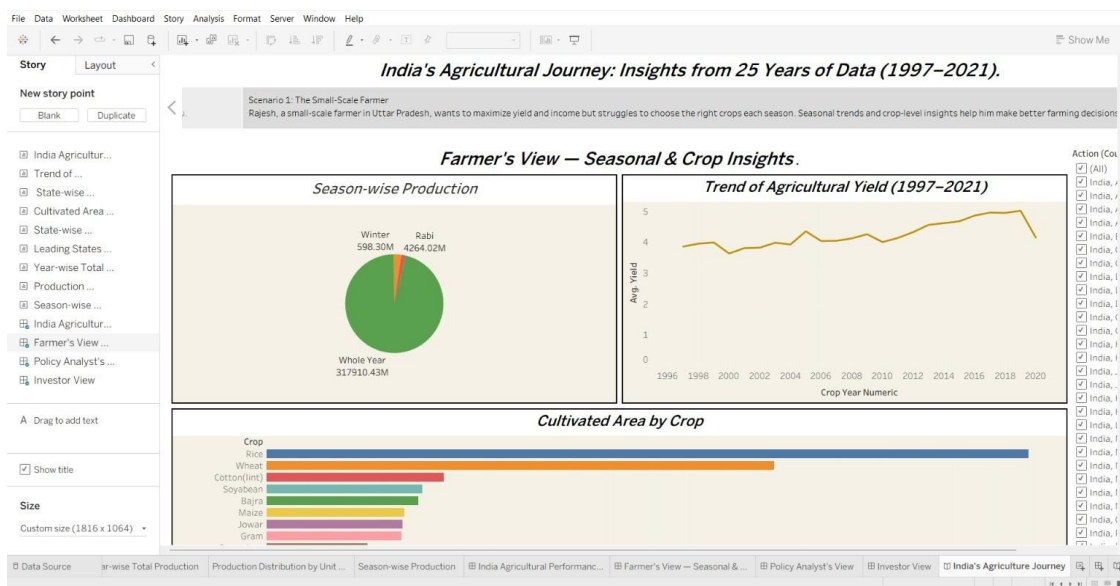
Dashboard 3 — Policy Analyst's View — Regional Performance & Trends



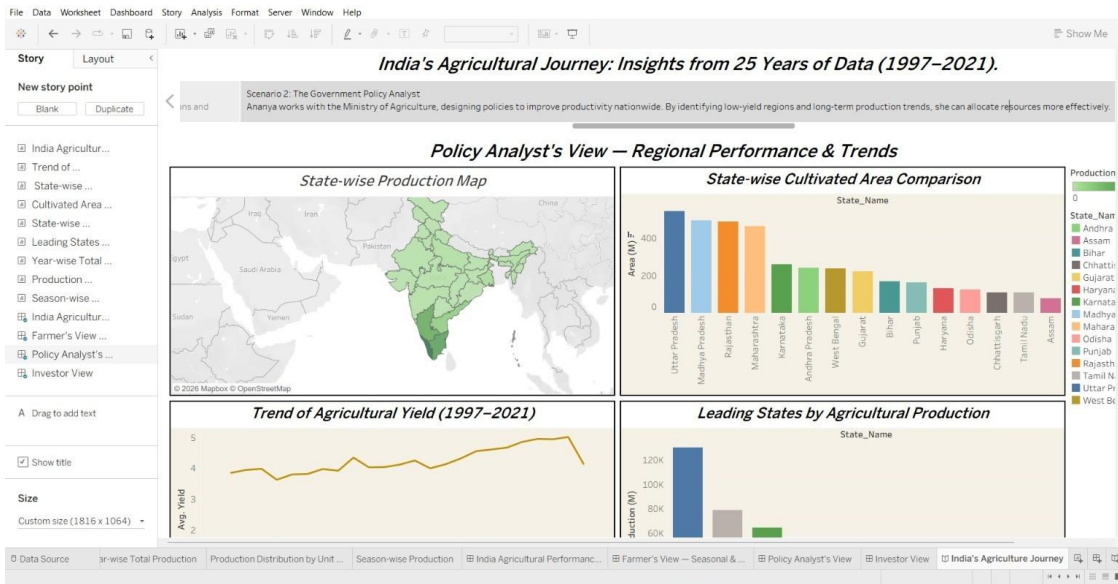
Dashboard 4 — Investor's View — Growth & Opportunity Insights



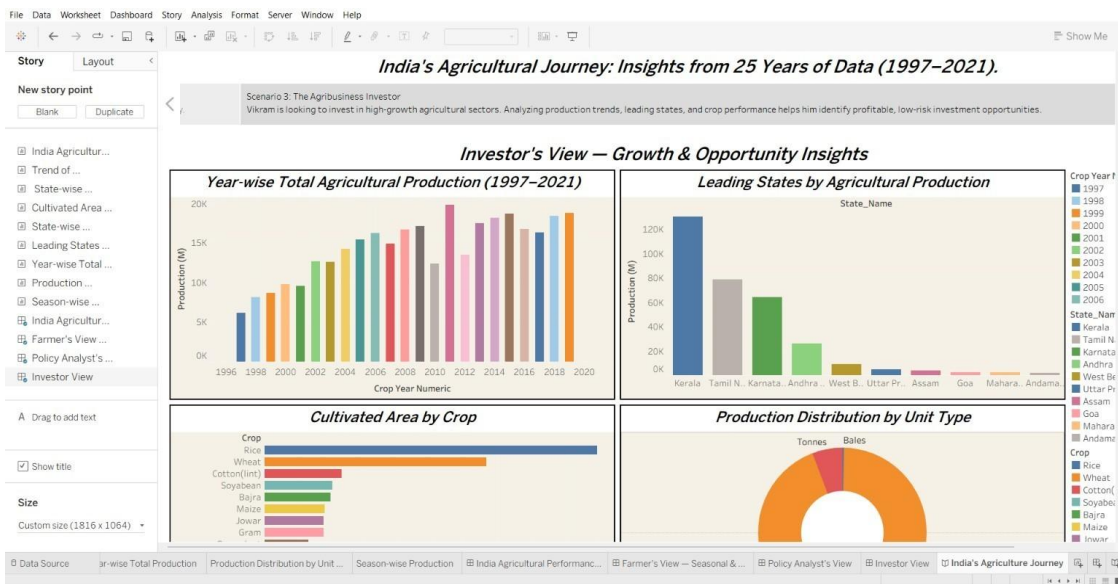
Story Point 1 — Overview



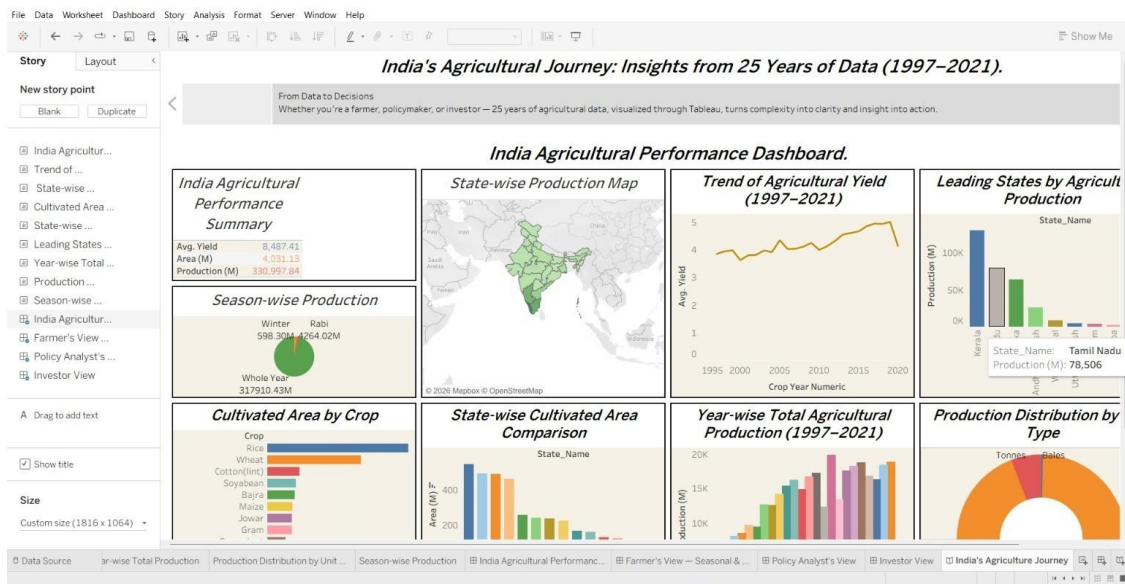
Story Point 2 — Scenario 1: The Small-Scale Farmer



Story Point 3 — Scenario 2: The Government Policy Analyst



Story Point 4 — Scenario 3: The Agribusiness Investor



Story Point 5 — From Data to Decisions

8. Advantages & Disadvantages

Advantages

- Zero-cost deployment: every tool in the stack (Python, Tableau Public, GitHub) is free, with no server, license, or IT approval required.
- Role-specific dashboards mean each persona reaches a relevant view directly rather than filtering a generic report from scratch.
- Sub-3-second filter response on a 345,407-row dataset, confirmed by Performance Testing, with no extraction or aggregation needed.
- No login barrier: the public Tableau link is accessible on any browser, which matters for low-digital-literacy users like Rajesh.
- Cross-browser and mobile-view compatibility, validated in Chrome, Edge, and Safari.
- Clear traceability from persona pain points through requirements to shipped charts, keeping the workbook focused rather than feature-bloated.

Disadvantages

- 2021 figures are partial-year, since year-end government reporting was incomplete at the time of data collection — visible as a dip at the end of the yield trend line rather than an actual decline.
- Production_Units differ by crop (Tonnes, Bales, Nuts); raw production rankings can be misleading unless read alongside the Production Distribution by Unit Type chart.
- District-level data exists in the schema but was not used for district-level charts in this phase, limiting drilldown granularity for a farmer wanting hyper-local detail.
- Tableau Public workbooks are visible to anyone with the link, which is appropriate here since the dataset is public, but would not suit a project involving sensitive data.
- As a solo-built project, there was no second reviewer or QA pass beyond the documented Performance Testing, so edge cases outside the eight tested scenarios are not formally verified.

9.Conclusion

This project successfully converts 25 years of raw, unfiltered Indian government crop production data into an interactive, role-aware Tableau Public workbook. By tracing every chart back to a specific persona pain point identified during Ideation, and validating every functional and non-functional requirement during Performance Testing, the finished solution gives a farmer, a policy analyst, and an investor each a dashboard suited to their own decision — without requiring any of them to install software, log in, or interpret raw spreadsheet data. All eight performance test cases passed with no defects, and the workbook, cleaning scripts, and full documentation set are version-controlled and publicly available via GitHub and Tableau Public.

10.Future Scope

- Extend the dataset with post-2021 years as government reporting is finalised, refreshing the existing workbook without redesigning any dashboard.
- Add district-level drill-down charts using the District_Name field already present in the schema but currently unused.
- Blend in rainfall and Minimum Support Price (MSP) data as new calculated fields to build a rainfall-adjusted yield metric and a price-vs-production view for investors.
- Introduce a farmer feedback loop to track whether data-informed crop choices actually improved outcomes across subsequent seasons.
- Migrate from Tableau Public to Tableau Server or Tableau Cloud if the project moves from an individual capstone to an institutional or government-hosted deployment, enabling authenticated, higher-frequency data refreshes.

11. Appendix

Source Code

The Python (Pandas) preprocessing scripts used to clean and enrich the raw government extract into skillwallet_crop_cleaned.csv are maintained in the GitHub repository listed below.

Dataset Link

Historical Indian Crop Production Data, 1997–2021 (government agricultural statistics) — cleaned extract: skillwallet_crop_cleaned.csv, 345,407 rows, 12 columns, covering 36 states/union territories, 57 crop types, and 6 growing seasons.

GitHub & Project Demo Link

- Tableau Public workbook (all dashboards + Story):
<https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject/IndiasAgricultureJourney?publish=yes>
- GitHub repository (cleaning scripts + documentation):
<https://github.com/divyakauluri-spec/India-sagricultural-crop-production-analysis>